

Name: _____

Teacher: _____



GOSFORD HIGH SCHOOL

2013

Year 12 HSC Mathematics

Assessment Task #2

INSTRUCTIONS:

TIME: 90 minutes + 5 minutes reading time

- Write using black or blue pen.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- In questions 8 – 11, show relevant mathematical reasoning and/or calculations

Questions 1 – 7	Multiple Choice	/7
Question 8	Trigonometric Functions	/12
Question 9	Integration	/12
Question 10	Integration	/12
Question 11	Trigonometric Functions	/12
TOTAL		/55

2

SECTION 1

7 Marks

Attempt Questions 1 – 7 on the multiple choice sheet provided.
Allow about 10 minutes for this section.

Question 1:

In evaluating a definite integral, a working step is shown by a student to be:

$$[x + \tan x]_0^{\frac{\pi}{8}}$$

The answer to 2 decimal places is:

- A. 0.39
- B. 0.40
- C. 0.80
- D. 0.81

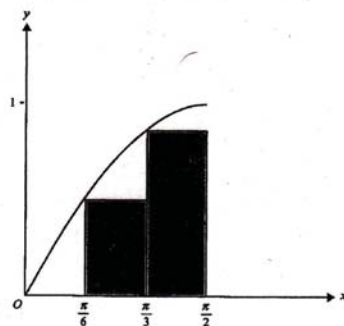
Questions 2:

What is the greatest value of the function

$$y = 3 - 2\cos x$$

- A. 1
- B. 3
- C. 5
- D. 6

Question 3:



The area under the curve $y = \sin x$ between $x = \frac{\pi}{6}$ and $x = \frac{\pi}{2}$ is approximated by 2 rectangles as shown above.

This approximation to the area is:

- A. $\frac{\sqrt{3}\pi}{6}$
- B. $\frac{\pi}{2}$
- C. $\frac{(\sqrt{3}+1)\pi}{12}$
- D. $\frac{(\sqrt{3}+1)\pi}{6}$

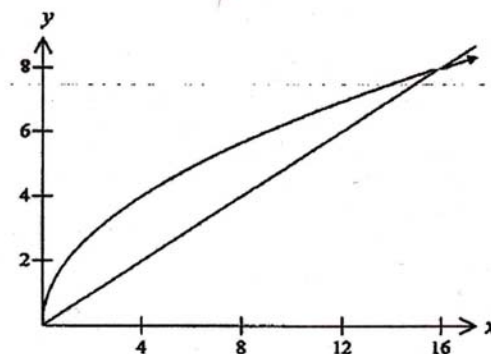
Question 4:

The function $f(x) = -3 \sin\left(\frac{\pi x}{5}\right)$ has a period of:

- A. 5
- B. 10
- C. $\frac{\pi}{5}$
- D. $\frac{\pi}{10}$

Question 5:

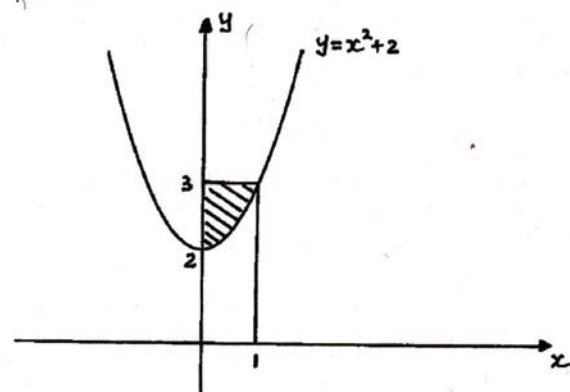
The diagram below shows the graph of $y = 2\sqrt{x}$ and $y = \frac{x}{4}$



Which of the following is the correct expression for the volume of the solid of revolution when the area between the curve $y = 2\sqrt{x}$ and $y = \frac{x}{4}$ is rotated around the x -axis?

- (A) $V = \int_0^8 (4y - \frac{y^2}{2}) dy$
- (B) $V = \int_0^{16} (2\sqrt{x} - \frac{x}{4}) dx$
- (C) $V = \pi \int_0^8 (16y^2 - \frac{y^4}{4}) dy$
- (D) $V = \pi \int_0^{16} (4x - \frac{x^2}{16}) dx$

Question 6:



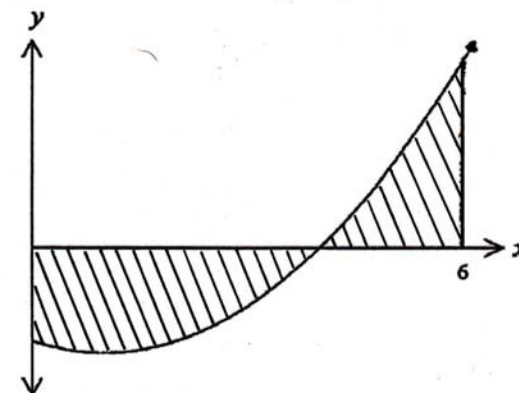
To find the area of the shaded region above, four different students proposed the following working:

- i. $\int_0^1 x^2 + 2 \, dx$
- ii. $3 - \int_0^1 x^2 + 2 \, dx$
- iii. $\int_2^3 \sqrt{y-2} \, dy$
- iv. $3 - \int_2^3 \sqrt{y-2} \, dy$

Which of the following is correct?

- A. (ii) only
- B. (ii) and (iii) only
- C. (i) and (ii) only
- D. (ii), (iii) and (iv) only

Question 7:



What is the correct expression for the area bounded by the x -axis and the curve $y = x^2 - 2x - 8$ between $0 \leq x \leq 6$? (As shown in the diagram above.)

- A. $A = \int_0^5 x^2 - 2x - 8 \, dx + \left| \int_5^6 x^2 - 2x - 8 \, dx \right|$
- B. $A = \int_0^4 x^2 - 2x - 8 \, dx + \left| \int_4^6 x^2 - 2x - 8 \, dx \right|$
- C. $A = \left| \int_0^5 x^2 - 2x - 8 \, dx \right| + \int_5^6 x^2 - 2x - 8 \, dx$
- D. $A = \left| \int_0^4 x^2 - 2x - 8 \, dx \right| + \int_4^6 x^2 - 2x - 8 \, dx$

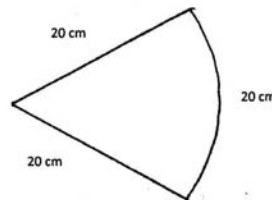
SECTION 2

48 Marks

- Attempt Questions 8 – 11 on your own writing paper, starting **each** question on a new sheet of paper.
- Allow about 1 hour and 20 minutes for this section.
- In Questions 8 – 11, your responses should include relevant mathematical reasoning and/or calculations.

Question 8: (12 marks) use a new sheet of writing paper.

- a. Convert 1.25 radians into degrees and minutes. (*Answer to the nearest minute*) 1
- b. Find the exact value of $\operatorname{cosec} \frac{4\pi}{3}$ 1
- c. Solve: $\sqrt{3} \tan x = 1$,
in the domain $-\pi \leq x \leq \pi$ 2
- d. Differentiate: 2
- i. $3 \sin(2 - 7x)$ 1
- ii. $x^2 \tan x$ 1
- iii. $\frac{\sin x}{1 + \cos x}$ 2
- e. Find $\int_0^{\frac{\pi}{2}} 4 \sec^2\left(\frac{x}{2}\right) dx$ 2
- f. The following sector: 2



Is cut from a circular piece of plastic and is **discarded**. Find the exact area of the **remaining** plastic.

Question 9: (12 Marks) Use a new sheet of writing paper.

- a. Find a primitive of: 1
- i. $6x^3 - 5x + 1$ 1
- ii. $(3x + 4)^8$ 1
- b. Find: $\int \frac{x^{-1/2}}{4} dx$ 1
- c. Evaluate: 2
- i. $\int_1^2 \frac{x^4 - x^2 + 3}{3x^2} dx$ 2
- ii. $\int_{-1}^0 \sqrt{1 - 2x} dx$ (*Leave in simplest exact form.*) 2
- d. The table below gives values for $y = f(x)$
- | | | | | |
|--------|---|-----|------|------|
| x | 0 | 1 | 2 | 3 |
| $f(x)$ | 2 | 1.5 | 1.41 | 1.37 |
- Using the four function values in the table and the trapezoidal rule approximate $\int_0^3 f(x) dx$, correct to 1 decimal place. 2
- e. Using Simpson's Rule with four strips find an approximate value for: 3

$$\int_0^2 \frac{4}{x^2 + 1} dx$$

(*Answer correct to 3 significant figures.*)

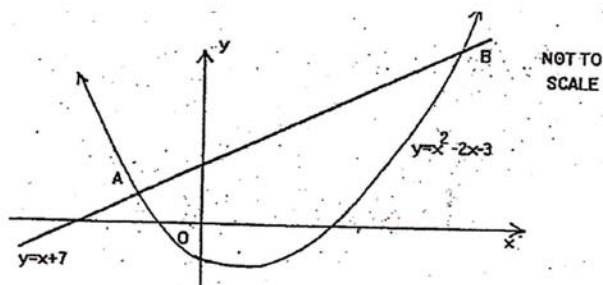
Question 10: (12 Marks) Use a new sheet of writing paper.

- a. The gradient of a curve is given by

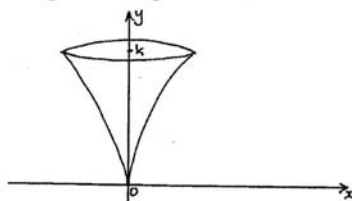
$$\frac{dy}{dx} = 2x - 3$$

Find the equation of the curve if it passes through the point (1,2)

- b. The area under the curve $y = (x-1)^2$ and between the ordinates $x = 1$, $x = 3$ and the x axis is rotated about the x axis. Find the exact volume of the solid of revolution formed.
- c. The diagram shows the graphs $y = x^2 - 2x - 3$ and $y = x + 7$. The graphs intersect at the points A and B.



- i. Find the coordinates of A and B.
- ii. Find the area enclosed by $y = x^2 - 2x - 3$ and $y = x + 7$
- d. The vase represented in the diagram is designed to hold $50\pi \text{ cm}^3$ of water when full.



Its shape is determined by rotating the parabola $x = \frac{y^2}{30}$ about the y axis. If the depth of water in the vase is k centimetres:

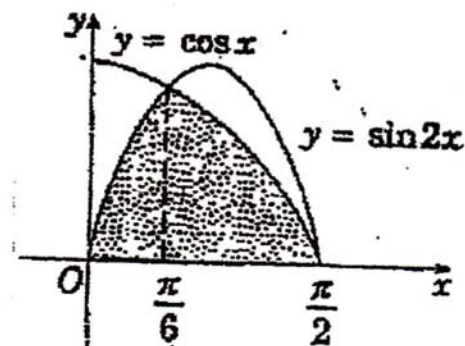
- i. Write an expression for this volume and hence show that

$$\int_0^k y^4 dy = 45\,000$$

- ii. Find the value of k , correct to 1 decimal place.

Question 11: (12 Marks) Use a new sheet of writing paper.

- a.



The diagram shows the graphs of the functions $y = \cos x$ and $y = \sin 2x$ between $x = 0$ and $x = \frac{\pi}{2}$

Calculate the area of the shaded region.

- b. On a $\frac{1}{3}$ page number plane, draw a neat sketch of $y = \sin(x + \frac{\pi}{4})$ in the domain $0 \leq x \leq 2\pi$.

(Clearly indicate and label all important features.)

- c. i. Show that $\sin \theta \tan \theta = \sec \theta - \cos \theta$
- ii. Hence, solve $\sin \theta \tan \theta = 0$ for $0 \leq \theta \leq 2\pi$
- d. i. Differentiate $\sin^3 4x$
- ii. Hence, evaluate $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \cos 4x - \cos^3 4x dx$

(Leave answer in exact form.)

END OF ASSESSMENT TASK

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SECTION 1

1/ D

$$\left[x + \tan x \right]_0^{\frac{\pi}{8}} = \left(\frac{\pi}{8} + \tan \frac{\pi}{8} \right) - 0$$

Using calculator and RAD. MODE

$$\text{answer} = 0.8069126.. \text{ (calc)}$$

$$= 0.81 \text{ (2d.p)}$$

2/ C

Since $-1 \leq \cos x \leq 1$, the maximum value of $3 - 2\cos x$ will occur when $\cos x = -1$

$$\text{i.e. } 3 - 2(-1) = 5$$

3/ C

The area of the 2 rectangles is given by

$$\text{Area} = \frac{\pi}{6} \times \sin \frac{\pi}{6} + \frac{\pi}{6} \times \sin \frac{\pi}{3}$$

$$= \frac{\pi}{6} \times \frac{1}{2} + \frac{\pi}{6} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{6} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right)$$

$$= \frac{(\sqrt{3} + 1)\pi}{12}$$

4/ B

The period of $y = a \sin nx$ is $\frac{2\pi}{n}$

$$\therefore \text{Period} = \frac{2\pi}{\frac{\pi}{5}} = 10$$

5/ D

$$\begin{aligned} \text{Using } V &= \pi \int_0^{16} y_1^2 - y_2^2 dx \\ &= \pi \int_0^{16} 4x - \frac{x^2}{16} dx \end{aligned}$$

6/ B

(i) gives a correct answer

(i.e. area of rectangle - area under the curve)

(ii) gives a correct answer

(i.e. finding the area between the curve, the y axis and $y=2, y=3$.)

7/ D

x intercepts of $y = x^2 - 2x - 8$ are $x = 4, -2$

$\therefore$ missing intercept in diagram is 4 and area is to be calculated

$$\left| \int_0^4 \right| + \int_4^6 \therefore D$$

SECTION 2

Q8, a) $1.25 \times \frac{180}{\pi}$
 $= 71.61972439..$ (calc)
 $= 71^{\circ} 37'$ (nearest min.)

b) $\operatorname{cosec} \frac{4\pi}{3} = -\frac{2}{\sqrt{3}}$

c) $\sqrt{3} \tan x = 1$
 $\tan x = \frac{1}{\sqrt{3}}$

for domain $0 \leq x \leq 2\pi$:

$$x = \frac{\pi}{6}, \frac{7\pi}{6}$$

for domain $-\pi \leq x \leq \pi$

$$x = \frac{\pi}{6}, -\frac{5\pi}{6}$$

d) (i) let $y = 3 \sin(2-7x)$
 $y' = -21 \cos(2-7x)$

(ii) let $y = x^2 \tan x$

(Product Rule) $y' = \tan x \cdot 2x + x^2 \sec^2 x$
 $= 2x \tan x + x^2 \sec^2 x$

(iii) let $y = \frac{\sin x}{1+\cos x}$

(Using quotient rule)

$$y' = \frac{(1+\cos x) \cdot \cos x - \sin x (-\sin x)}{(1+\cos x)^2}$$

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2}$$

$$= \frac{\cos x + 1}{(1+\cos x)^2}$$

$$= \frac{1}{1+\cos x}$$

e) $\int_0^{\frac{\pi}{2}} 4 \sec^2 \frac{x}{2} dx$
 $= \left[8 \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}}$

$$= 8 \tan \frac{\pi}{4} - 0$$

$$= 8$$

f) Area of remaining plastic

= Area of circle - Area of Sector

$$= \pi r^2 - \frac{1}{2} r^2 \theta$$

$$= \pi \times 20^2 - \frac{1}{2} \times 20^2 \times 1$$

$$= 400\pi - 200 \text{ cm}^2$$

Q9

a) (i) $\frac{4x^4}{4} - \frac{5x^2}{2} + x$

$$= \frac{3x^4}{2} - \frac{5x^2}{2} + x$$

(ii) $\frac{(3x+4)^9}{3 \times 9}$
 $= \frac{(3x+4)^9}{27}$

b) $\int \frac{x^{-1/2}}{4} dx = \frac{x^{1/2}}{2} + c$

(this form is consistent with question)

does not need:

$$\frac{\sqrt{x}}{2} + c$$

c) $\int_1^2 \frac{x^4 - x^2 + 3}{3x^2} dx$

$$= \int_1^2 \left(\frac{x^2}{3} - \frac{1}{3} + x^{-2} \right) dx$$

$$= \left[\frac{x^3}{9} - \frac{x}{3} - x^{-1} \right]_1^2$$

$$= \left(\frac{8}{9} - \frac{2}{3} - \frac{1}{2} \right) - \left(\frac{1}{9} - \frac{1}{3} - 1 \right)$$

$$= \frac{17}{18}$$

(ii) $\int_{-1}^0 \sqrt{1-2x} dx$

$$= \int_{-1}^0 (1-2x)^{1/2} dx$$

$$= \left[\frac{(1-2x)^{3/2}}{3/2 \times 2} \right]_{-1}^0$$

$$= -\frac{1}{3} \left[\sqrt{1-2x}^3 \right]_{-1}^0$$

$$= -\frac{1}{3} (1 - \sqrt{27})$$

$$= -\frac{1}{3} + \sqrt{3}$$

d) Using $\frac{1}{2} [y_0 + y_3 + 2(y_1 + y_2)]$

$$= \frac{1}{2} [2 + 1.37 + 2(1.5 + 1.41)]$$

$$= 4.595$$

$$= 4.6 \text{ (1 d.p.)}$$

e) 4 strips $\rightarrow$ 5 function values

x	0	0.5	1	1.5	2
y	4	3.2	2	1.231	0.8

$y_0 \ y_1 \ y_2 \ y_3 \ y_4$

$$\int_0^2 \frac{4}{x^2+1} dx \approx \frac{1}{3} [y_0 + y_4 + 4(y_1 + y_3) + 2y_2]$$

$$= \frac{0.5}{3} [4 + 0.8 + 4(3.2 + 1.231) + 2 \times 2]$$

$$= 4.42 \text{ (3 s.f.)}$$

Q10

$$a) \frac{dy}{dx} = 2x - 3$$

$$y = x^2 - 3x + c$$

$$\text{Sub (1,2)} : 2 = 1 - 3 + c$$

$$2 = -2 + c$$

$$4 = c$$

$\therefore$ Eqn of curve is :

$$x^2 - 3x + 4$$

$$b) V = \pi \int_1^3 (x-1)^4 dx$$

$$= \frac{\pi}{5} [(x-1)^5]_1^3$$

$$= \frac{\pi}{5} (32 - 0)$$

$$= \frac{32\pi}{5} \text{ units}^3$$

$$c) (1) y = x^2 - 2x - 3 \quad - (1)$$

$$y = x + 7 \quad - (2)$$

Sub (1) in (2)

$$x^2 - 2x - 3 = x + 7$$

$$x^2 - 3x - 10 = 0$$

$$(x-5)(x+2) = 0$$

$$x = 5, -2$$

$$y = 12, 5$$

$$\therefore A = (-2, 5) \quad B = (5, 12)$$

$$(ii) \text{ Area} = \int_{-2}^5 (x+7) - (x^2 - 2x - 3) dx$$

$$= \int_{-2}^5 3x - x^2 + 10 dx$$

$$= \left[\frac{3x^2}{2} - \frac{x^3}{3} + 10x \right]_{-2}^5$$

$$= \left(\frac{75}{2} - \frac{125}{3} + 50 \right) - \left(6 + \frac{8}{3} - 20 \right)$$

$$= 45\frac{5}{6} - (-11\frac{1}{3})$$

$$= 57\frac{1}{6} \text{ units}^2$$

$$d) (i) V = \pi \int_0^k x^2 dy$$

$$= \pi \int_0^k \frac{y^4}{900} dy$$

$$= \frac{\pi}{900} \int_0^k y^4 dy$$

$$\therefore 50\pi = \frac{\pi}{900} \int_0^k y^4 dy$$

$$\therefore \int_0^k y^4 dy = 45000 \text{ as req.}$$

$$(ii) \left[\frac{y^5}{5} \right]_0^k = 45000$$

$$\frac{k^5}{5} - 0 = 45000$$

$$k^5 = 225000$$

$$k = 11.8 \text{ (1 d.p.)}$$

Q11

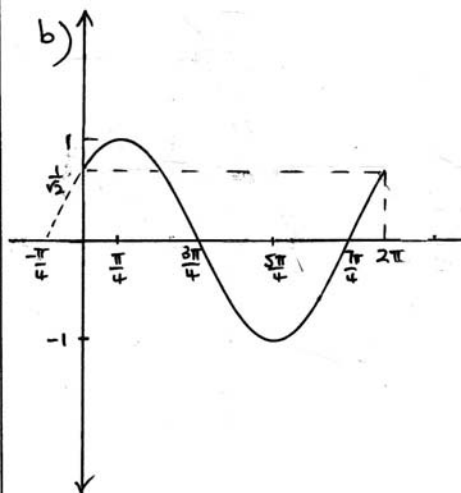
$$a) \text{ Area} = \int_0^{\frac{\pi}{6}} \sin 2x dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos x dx$$

$$= -\frac{1}{2} \left[\cos 2x \right]_0^{\frac{\pi}{6}} + \left[\sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= -\frac{1}{2} \left(\frac{1}{2} - 1 \right) + \left(1 - \frac{1}{2} \right)$$

$$= \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4} \text{ units}^2$$



$$c) (1) \text{ LHS} = \sin \theta \tan \theta$$

$$= \sin \theta \cdot \frac{\sin \theta}{\cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta}$$

$$\text{RHS} = \sec \theta - \cos \theta$$

$$= \frac{1}{\cos \theta} - \cos \theta$$

$$= \frac{1 - \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta}$$

$$= \text{LHS as req.}$$

$$(ii) \sin \theta \tan \theta = 0$$

$$\therefore \sec \theta - \cos \theta = 0$$

$$\frac{1}{\cos \theta} = \cos \theta$$

$$1 = \cos^2 \theta$$

$$\therefore \cos \theta = \pm 1$$

$$\theta = 0, \pi, 2\pi$$

$$d) (1) \text{ Let } y = \sin^3 4x$$

$$y' = 3 \sin^2 4x \times 4 \cos 4x$$

$$= 12 \sin^2 4x \cos 4x$$

$$= 12 (1 - \cos^2 4x) \cos 4x$$

$$= (12 - 12 \cos^2 4x) \cos 4x$$

$$= 12 (\cos 4x - \cos^3 4x)$$

$$\therefore \int_{\frac{\pi}{8}}^{\frac{\pi}{2}} \cos 4x - \cos^3 4x dx = \frac{1}{12} \left[\sin^3 4x \right]_{\frac{\pi}{8}}^{\frac{\pi}{2}}$$

$$= \frac{1}{12} \left(\sin^3 \frac{2\pi}{2} - \sin^3 \frac{\pi}{2} \right) = \frac{1}{12} \left(\frac{3\sqrt{3}}{8} - 1 \right)$$